Week 10-Count data

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2026-04-30

1 Poisson distribution

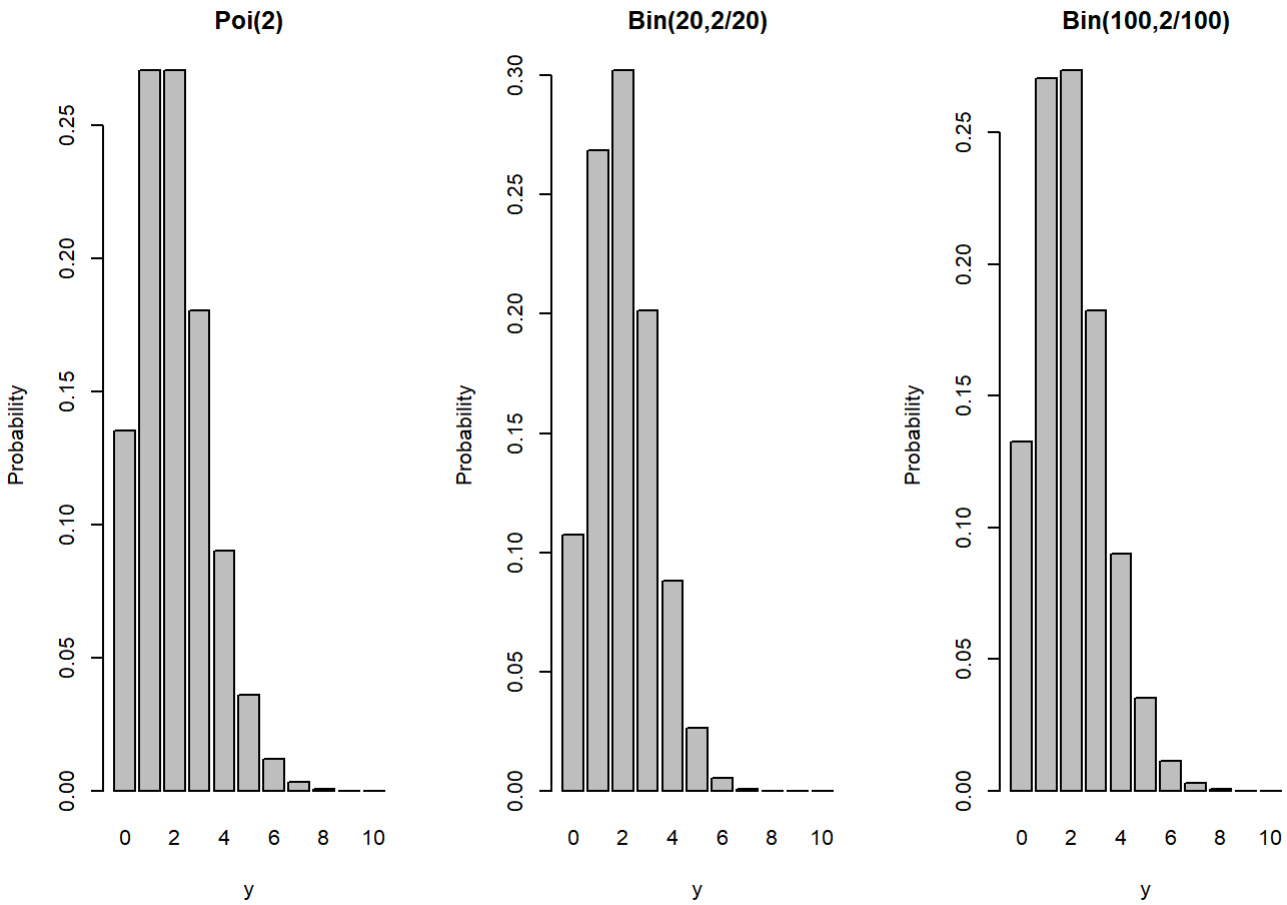
$$Y \sim Poisson(\mu), \quad E(Y) = \mu, \quad Var(Y) = \mu$$

$$P(Y = y) = \frac{e^{-\mu}\mu^y}{y!}, \quad y = 0, 1, 2, \ldots$$

1.1 Poisson vs Binomial

When you have a massive number of trials (n) but the probability of success (p) is tiny, the Binomial distribution converges toward a Poisson distribution. ($\mu = np$)

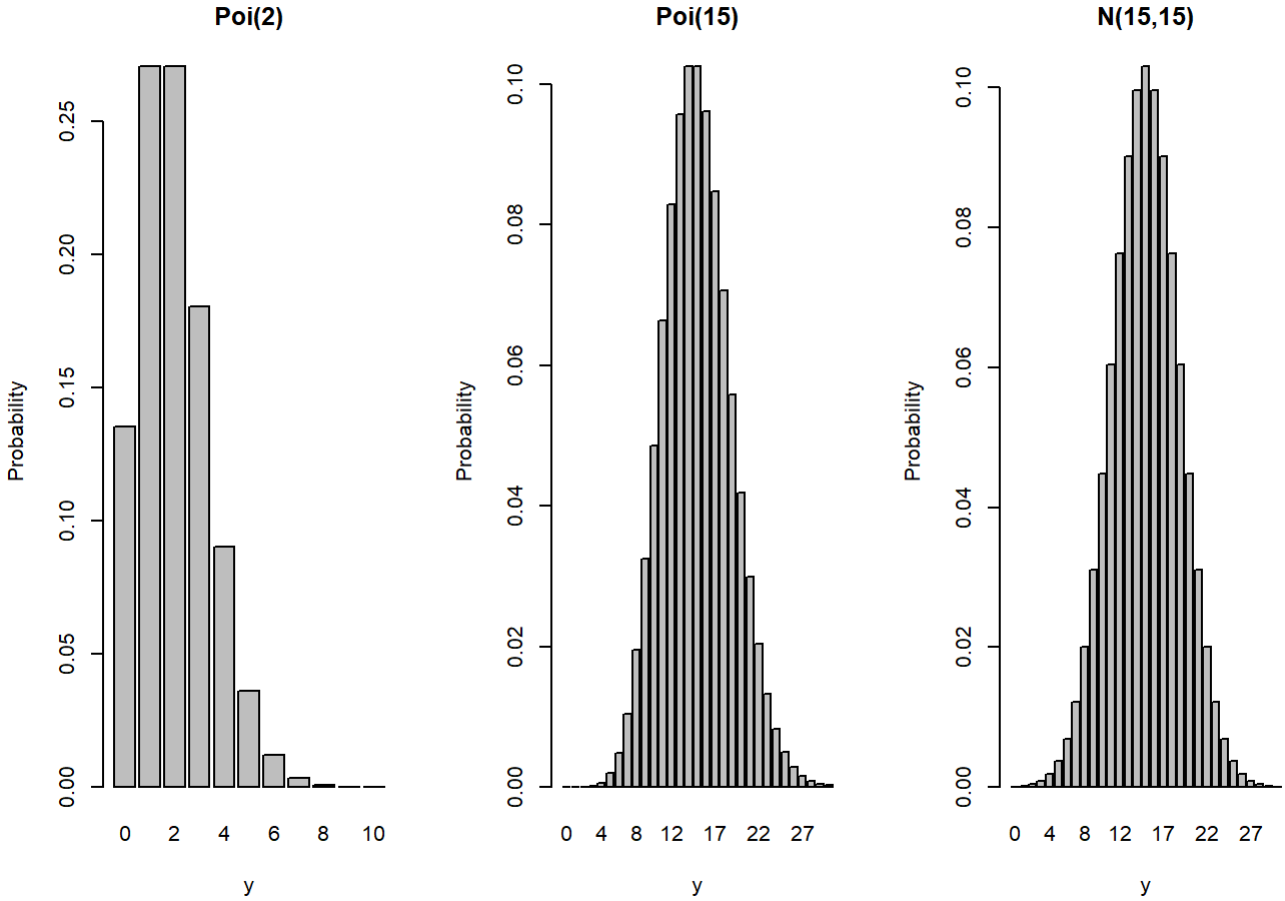
```
par(mfrow=c(1,3))
barplot(dpois(0:10,2),xlab="y",ylab="Probability",names=0:10,main="Poi(2)")
barplot(dbinom(0:10,10,2/10),xlab="y",ylab="Probability",names=0:10,main="Bin(20,2/20)")
barplot(dbinom(0:10,100,2/100),xlab="y",ylab="Probability",names=0:10,main="Bin(100,2/100)")
```



1.2 Poisson vs Normal

For a Poisson distribution, when μ is large, the distribution can be approximated by $N(\mu, \mu)$ (The “tail” on the right starts to balance out with the left side.)

```
par(mfrow=c(1,3))
barplot(dpois(0:10,2),xlab="y",ylab="Probability",names=0:10,main="Poi(2)")
barplot(dpois(0:30,15),xlab="y",ylab="Probability",names=0:30,main="Poi(15)")
barplot(dnorm(0:30,mean=15,sd=sqrt(15)), xlab="y",ylab="Probability",names=0:30,main="N(15,1
5)")
```



Example: Supermarket Checkout Counts

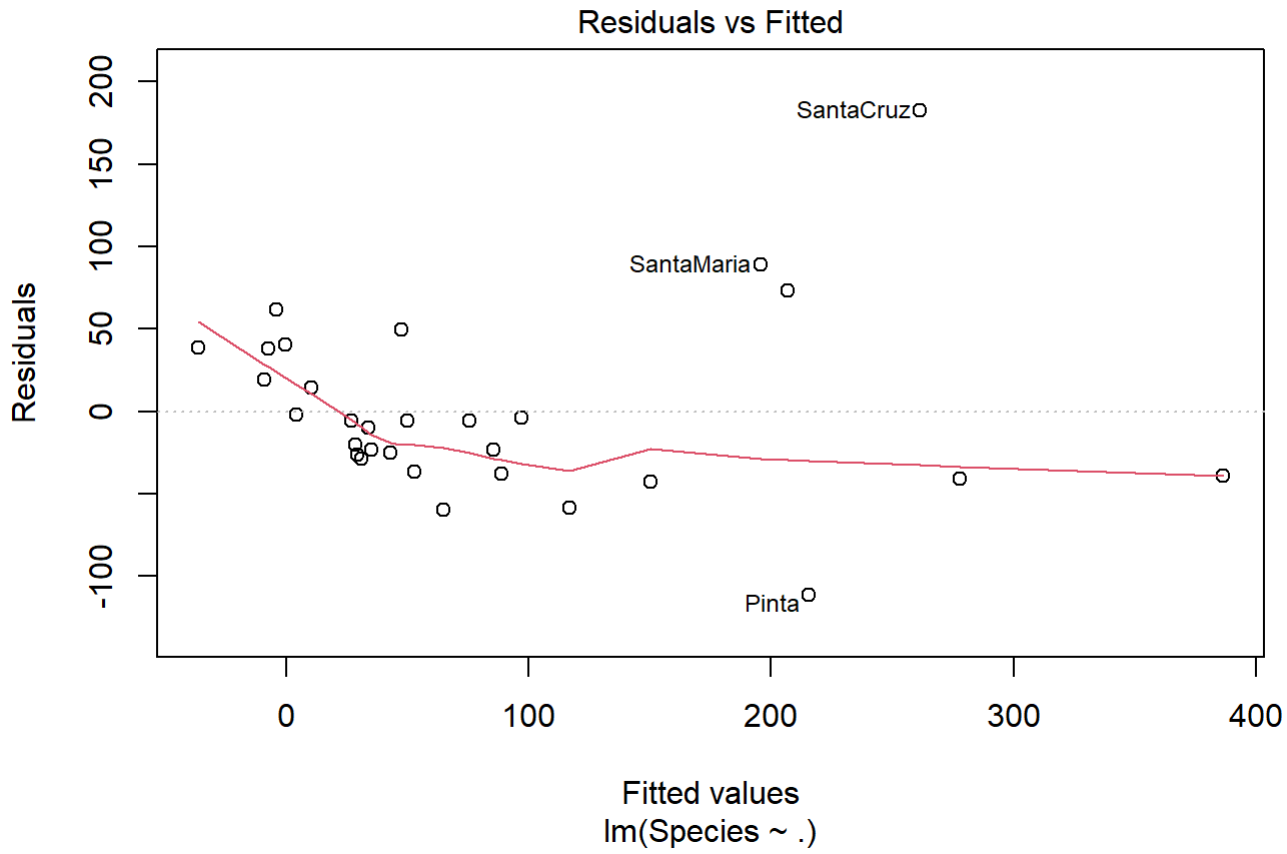
-Scenario 1 (Poisson): The supermarket during late-night hours, on average, only 2 customers ($\mu = 2$) check out per hour.

-Scenario 2 (Normal): The weekday morning, the average rises to 15 customers ($\mu = 15$) per hour.

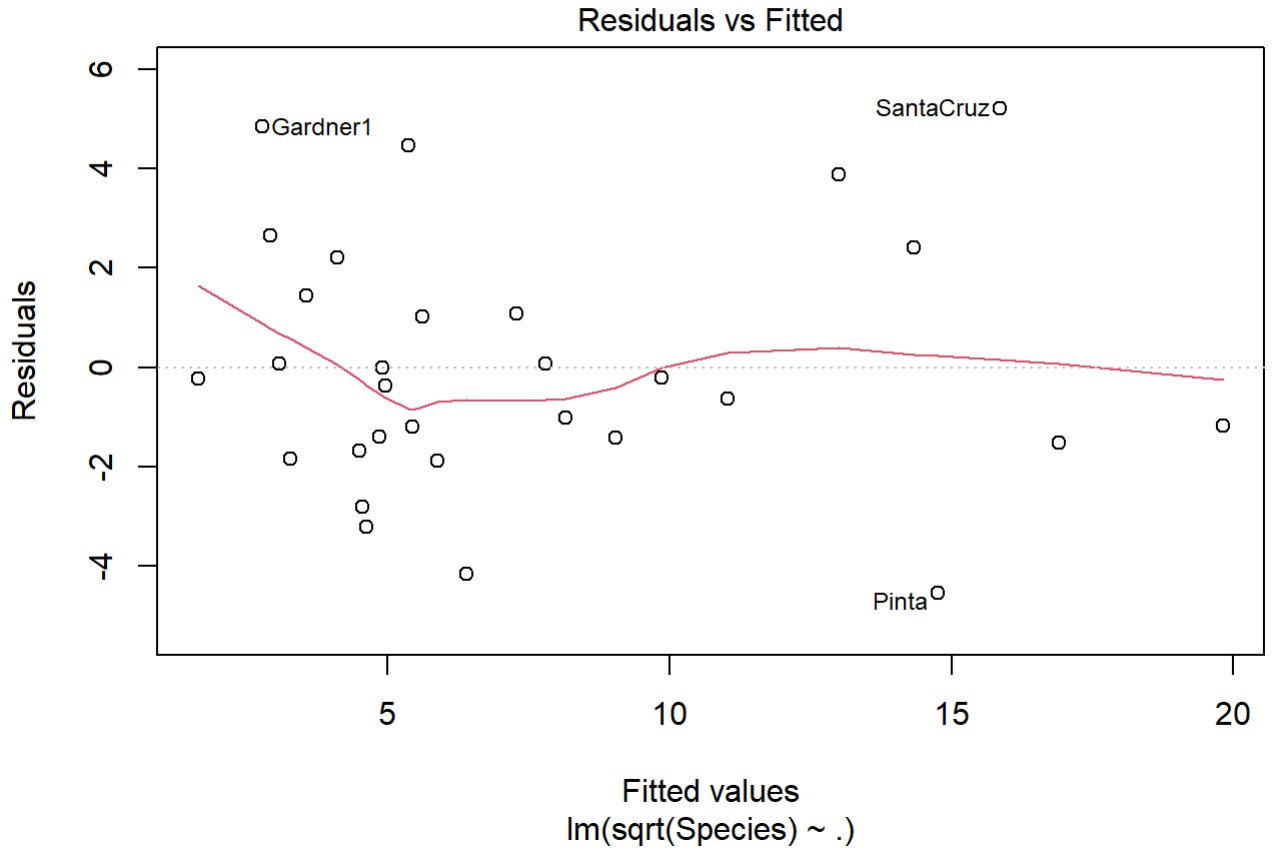
2 Poisson regression

```
## gala data
data(gala, package="faraway")
gala <- gala[,-2] #throw out the "Endemics" in this analysis

# fit linear model
modl <- lm(Species ~ . , gala)
plot(modl, 1) # unequal variance
```




```
modt <- lm(sqrt(Species) ~ . , gala)
plot(modt, 1)
```



```
summary(modt)
```

```
##
## Call:
## lm(formula = sqrt(Species) ~ ., data = gala)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.5572 -1.4969 -0.3031  1.3527  5.2110
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.3919243  0.8712678   3.893 0.000690 ***
## Area        -0.0019718  0.0010199  -1.933 0.065080 .
## Elevation    0.0164784  0.0024410   6.751 5.55e-07 ***
## Nearest      0.0249326  0.0479495   0.520 0.607844
## Scruz       -0.0134826  0.0097980  -1.376 0.181509
## Adjacent    -0.0033669  0.0008051  -4.182 0.000333 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.774 on 24 degrees of freedom
## Multiple R-squared:  0.7827, Adjusted R-squared:  0.7374
## F-statistic: 17.29 on 5 and 24 DF,  p-value: 2.874e-07
```

Mod1 showed non-constant variance, but the fit improved after the response was square-root transformed.

Fit the Poisson regression to the Galapagos data:

```
# fit poisson regression model
modp <- glm(Species ~ ., family=poisson, gala)
summary(modp)
```

```
##
## Call:
## glm(formula = Species ~ ., family = poisson, data = gala)
##
## Coefficients:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)  3.155e+00  5.175e-02  60.963  < 2e-16 ***
## Area        -5.799e-04  2.627e-05 -22.074  < 2e-16 ***
## Elevation    3.541e-03  8.741e-05  40.507  < 2e-16 ***
## Nearest      8.826e-03  1.821e-03   4.846 1.26e-06 ***
## Scruz       -5.709e-03  6.256e-04  -9.126  < 2e-16 ***
## Adjacent    -6.630e-04  2.933e-05 -22.608  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 3510.73  on 29  degrees of freedom
## Residual deviance:  716.85  on 24  degrees of freedom
## AIC: 889.68
##
## Number of Fisher Scoring iterations: 5
```

- Deviance (G-statistic) = 716.85

2.1 Goodness of fit test

$$H_0 : \text{The Poisson distribution is appropriate}$$

$$H_1 : \text{The model does not fit the data}$$

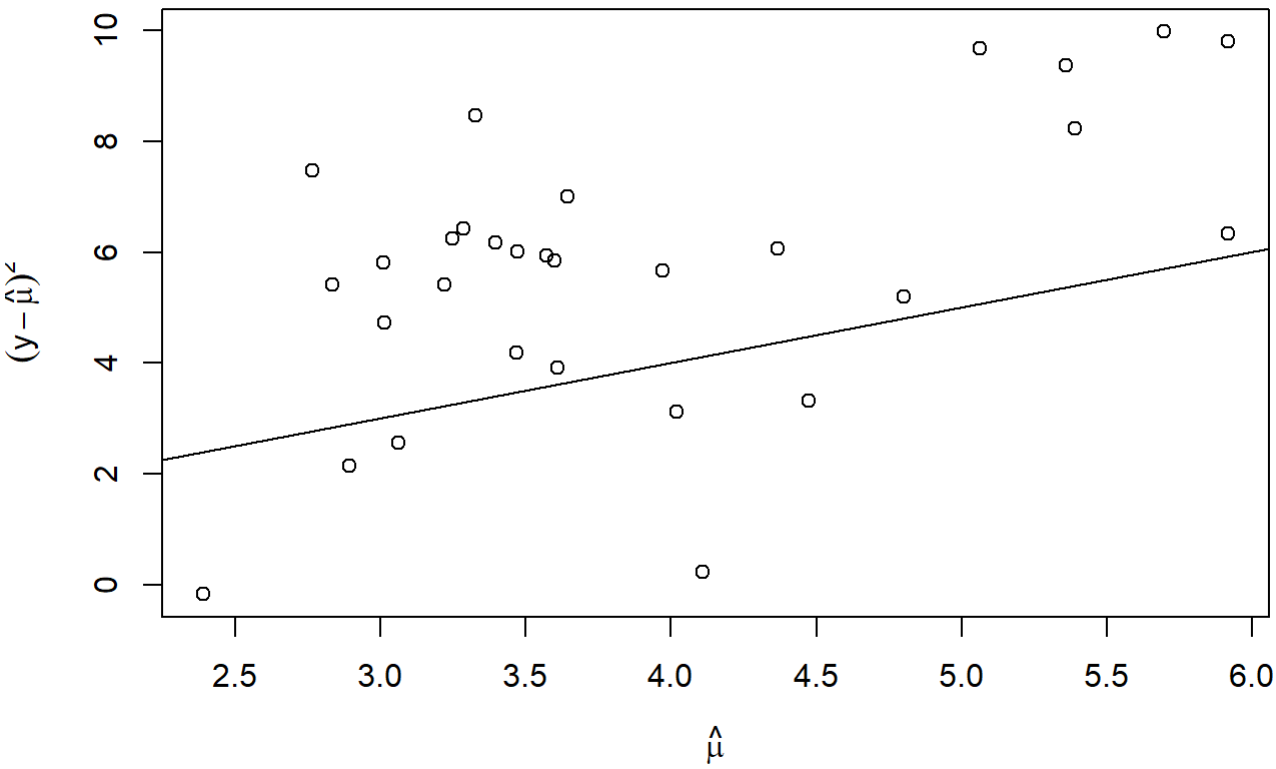
```
p_val <- 1 - pchisq(summary(modp)$deviance, summary(modp)$df.residual) #compare with saturate
d model
```

H_0: the model is appropriate; the model is not significantly different from the saturated model

Reject H_0 means that the model is “to far away” from the saturated model to be considered a good fit. In the context of Poisson regression, this “distance” is almost always cause by Overdispersion.

3 dispersed poisson model

```
## dispersed  poisson model
plot(log(fitted(modp)),log((gala$Species-fitted(modp))^2), xlab=expression(hat(mu)),ylab=expr
ession((y-hat(mu))^2))
abline(0,1)
```



The variance is proportional to, but larger than, the mean.

Suppose

$$Y \sim Poi(\lambda)$$

Let λ be gamma distributed with

$$E(\lambda) = \mu, \quad Var(\lambda) = \frac{\mu}{\phi}$$

Then

$$Var(Y) = \frac{\mu(1+\phi)}{\phi}$$

- $\phi = 1$ no overdispersion
- $\phi > 1$ overdispersion

The dispersion parameter may be estimated using:

$$\hat{\phi} = \frac{X^2}{n-p} = \frac{\sum_i (y_i - \hat{\mu}_i)^2 / \hat{\mu}_i}{n-p}$$

```
# estimate dispersion parameter
(dp <- sum(residuals(modp,type="pearson")^2)/modp$df.res)
```

```
## [1] 31.74914
```

```
# fit model with dispersion
summary(modp,dispersion=dp)
```

```
##
## Call:
## glm(formula = Species ~ ., family = poisson, data = gala)
##
## Coefficients:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)  3.1548079  0.2915897  10.819  < 2e-16 ***
## Area        -0.0005799  0.0001480  -3.918 8.95e-05 ***
## Elevation    0.0035406  0.0004925   7.189 6.53e-13 ***
## Nearest      0.0088256  0.0102621   0.860    0.390
## Scruz       -0.0057094  0.0035251  -1.620    0.105
## Adjacent    -0.0006630  0.0001653  -4.012 6.01e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 31.74914)
##
##     Null deviance: 3510.73  on 29  degrees of freedom
## Residual deviance:  716.85  on 24  degrees of freedom
## AIC: 889.68
##
## Number of Fisher Scoring iterations: 5
```

Changing the dispersion parameter does not shift the regression coefficients, but it does change their standard errors. This ensures that the resulting p-values and confidence intervals accurately account for the extra variation in the model.

4 Rate model

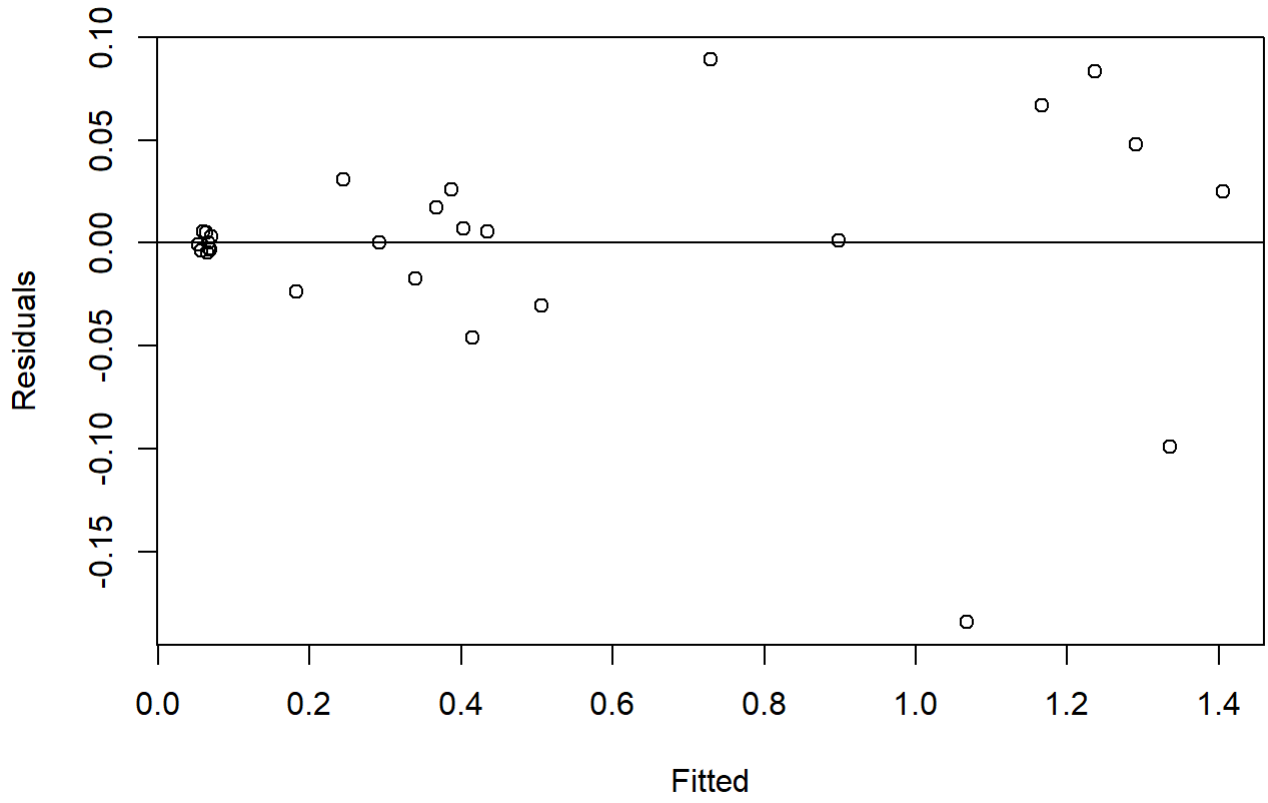
Determine the effect of gamma radiation on the numbers of chromosomal abnormalities (ca) observed. The number (cells), in hundreds of cells exposed in each run, differs. The dose amount (doseamt) and the rate (doserate) at which the dose is applied are the predictors of interest.

```
## rate model
data(dicentric, package="faraway")

# fit linear model: model the rate
lmod <- lm(ca/cells ~ log(doserate)*factor(doseamt), dicentric)
summary(lmod) #adjust R square: fit well
```

```
##
## Call:
## lm(formula = ca/cells ~ log(doserate) * factor(doseamt), data = dicentric)
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.184275 -0.004212  0.001314  0.021208  0.089076
##
## Coefficients:
##                                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)                      0.063489   0.019528   3.251  0.00382 **
## log(doserate)                    0.004573   0.016692   0.274  0.78680
## factor(doseamt)2.5               0.276315   0.027616  10.005 1.92e-09 ***
## factor(doseamt)5                 1.004119   0.027616  36.359  < 2e-16 ***
## log(doserate):factor(doseamt)2.5 0.063933   0.023606   2.708  0.01317 *
## log(doserate):factor(doseamt)5   0.239129   0.023606  10.130 1.54e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.05858 on 21 degrees of freedom
## Multiple R-squared:  0.9874, Adjusted R-squared:  0.9844
## F-statistic:   330 on 5 and 21 DF,  p-value: < 2.2e-16
```

```
# diagnosis of the linear model
plot(residuals(lmod) ~ fitted(lmod),xlab="Fitted",ylab="Residuals") # unequal variance
abline(h=0)
```



```
dicentric$dosef <- factor(dicentric$doseamt)
```

Try an approach that directly models the count response.

```
# fit poisson regression model
pmod <- glm(ca ~ log(cells)+log(doserate)*dosef,  family=poisson,dicentric)
summary(pmod)
```

```
##
## Call:
## glm(formula = ca ~ log(cells) + log(doserate) * dosef, family = poisson,
##     data = dicentric)
##
## Coefficients:
##                        Estimate Std. Error z value Pr(>|z|)
## (Intercept)            -2.76534    0.38116  -7.255 4.02e-13 ***
## log(cells)              1.00252    0.05137  19.517  < 2e-16 ***
## log(doserate)           0.07200    0.03547   2.030 0.042403 *
## dosef2.5                1.62984    0.10273  15.866  < 2e-16 ***
## dosef5                  2.76673    0.12287  22.517  < 2e-16 ***
## log(doserate):dosef2.5  0.16111    0.04837   3.331 0.000866 ***
## log(doserate):dosef5    0.19316    0.04299   4.493 7.03e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 916.127  on 26  degrees of freedom
## Residual deviance:  21.748  on 20  degrees of freedom
## AIC: 211.15
##
## Number of Fisher Scoring iterations: 4
```

We can see that the coefficient of 1.0025 is very close to one. This suggests fitting a model with the coefficient fixed as one.

$$\log(\frac{\text{ca}}{\text{cells}}) = X\beta$$

$$\log(\text{ca}) = \log(\text{cells}) + X\beta$$

```
# fit a rate model (set the coefficient of log(cells) = 1)
rmod <- glm(ca ~ offset(log(cells))+log(doserate)*dosef,  family=poisson, dicentric)
summary(rmod)
```

```
##
## Call:
## glm(formula = ca ~ offset(log(cells)) + log(doserate) * dosef,
##     family = poisson, data = dicentric)
##
## Coefficients:
##                        Estimate Std. Error z value Pr(>|z|)
## (Intercept)            -2.74671    0.03426 -80.165  < 2e-16 ***
## log(doserate)           0.07178    0.03518   2.041 0.041299 *
## dosef2.5                1.62542    0.04946  32.863  < 2e-16 ***
## dosef5                  2.76109    0.04349  63.491  < 2e-16 ***
## log(doserate):dosef2.5  0.16122    0.04830   3.338 0.000844 ***
## log(doserate):dosef5    0.19350    0.04243   4.561  5.1e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 4753.00  on 26  degrees of freedom
## Residual deviance:   21.75  on 21  degrees of freedom
## AIC: 209.16
##
## Number of Fisher Scoring iterations: 4
```